COMPUTER ENGINEERING

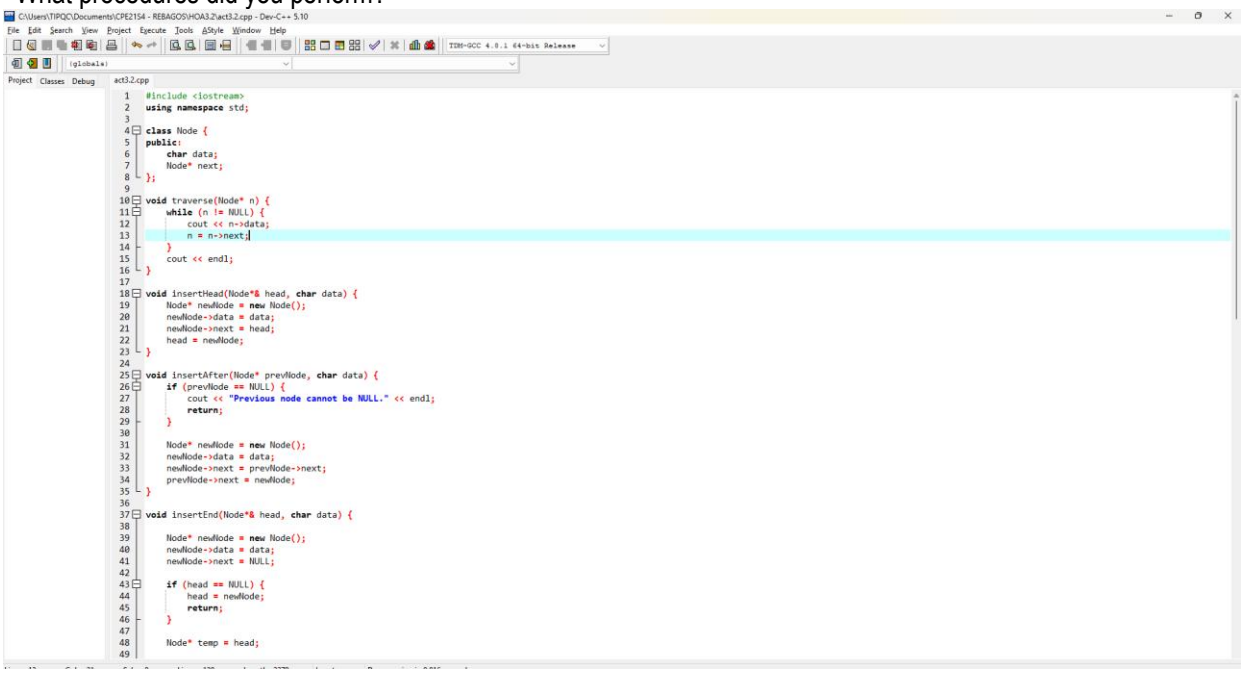
QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S4
DATE PERFORMED:	07/29/26	**DATE SUBMITTED:**	07/29/26
NAME:	KIAN JOROS C. REBAGOS	**INSTRUCTOR:**	Engr. Jimlord Quejado

ACTIVITY NO. ___ [ACTIVITY NAME]

1. PROCEDURE OUTPUT

What procedures did you perform?



```
#include <iostream>
using namespace std;

class Node {
public:
    char data;
    Node* next;
};

void traverse(Node* n) {
    while (n != NULL) {
        cout << n->data;
        n = n->next;
    }
    cout << endl;
}

void insertHead(Node*& head, char data) {
    Node* newNode = new Node();
    newNode->data = data;
    newNode->next = head;
    head = newNode;
}

void insertAfter(Node* prevNode, char data) {
    if (prevNode == NULL) {
        cout << "Previous node cannot be NULL." << endl;
        return;
    }

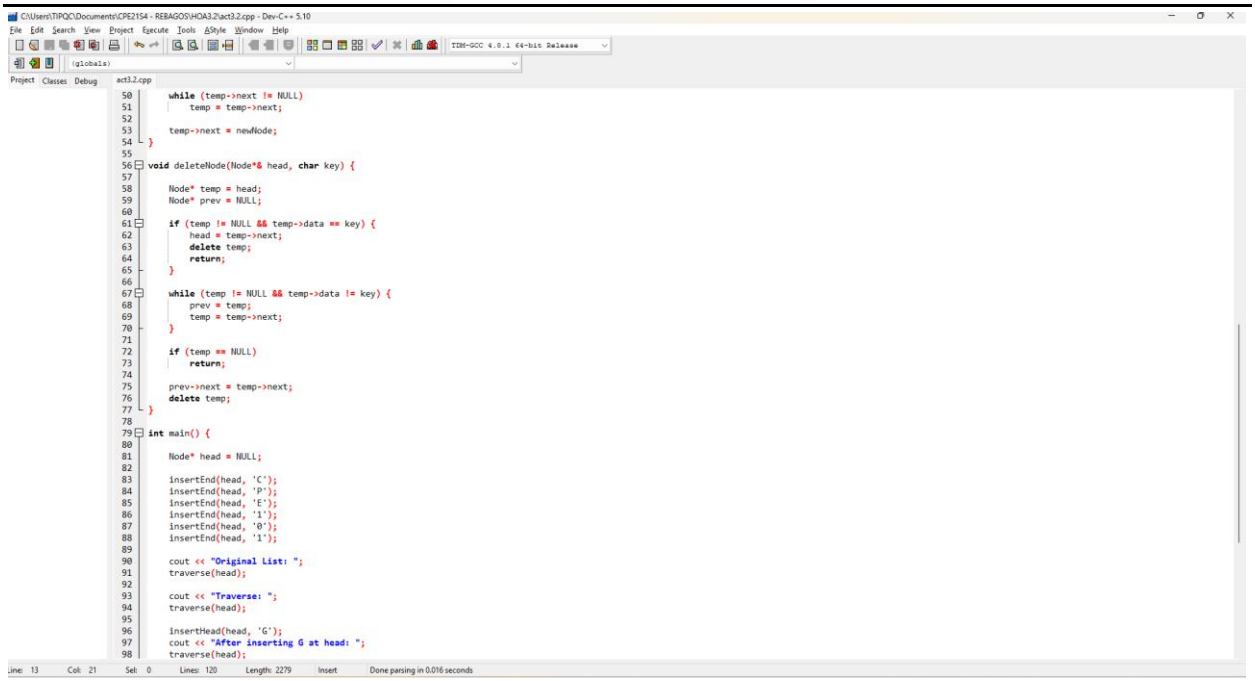
    Node* newNode = new Node();
    newNode->data = data;
    newNode->next = prevNode->next;
    prevNode->next = newNode;
}

void insertEnd(Node*& head, char data) {

    Node* newNode = new Node();
    newNode->data = data;
    newNode->next = NULL;

    if (head == NULL) {
        head = newNode;
        return;
    }

    Node* temp = head;
```



```
    while (temp->next != NULL)
        temp = temp->next;

    temp->next = newNode;
}

void deleteNode(Node*& head, char key) {

    Node* temp = head;
    Node* prev = NULL;

    if (temp != NULL && temp->data == key) {
        head = temp->next;
        delete temp;
        return;
    }

    while (temp != NULL && temp->data != key) {
        prev = temp;
        temp = temp->next;
    }

    if (temp == NULL)
        return;

    prev->next = temp->next;
    delete temp;
}

int main() {

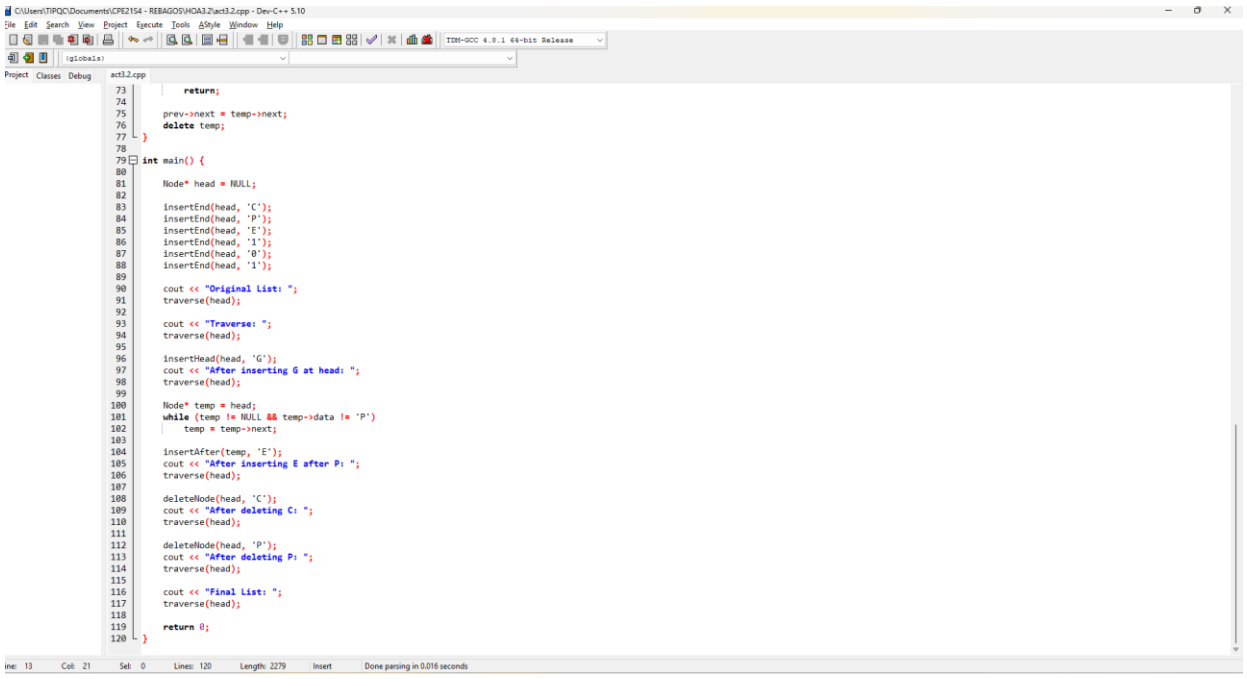
    Node* head = NULL;

    insertEnd(head, 'C');
    insertEnd(head, 'P');
    insertEnd(head, 'E');
    insertEnd(head, '1');
    insertEnd(head, '0');
    insertEnd(head, '1');

    cout << "Original List: ";
    traverse(head);

    cout << "Traverse: ";
    traverse(head);

    insertHead(head, 'G');
    cout << "After inserting G at head: ";
    traverse(head);
```



```
        return;

    prev->next = temp->next;
    delete temp;
}

int main() {

    Node* head = NULL;

    insertEnd(head, 'C');
    insertEnd(head, 'P');
    insertEnd(head, 'E');
    insertEnd(head, '1');
    insertEnd(head, '0');
    insertEnd(head, '1');

    cout << "Original List: ";
    traverse(head);

    cout << "Traverse: ";
    traverse(head);

    insertHead(head, 'G');
    cout << "After inserting G at head: ";
    traverse(head);

    Node* temp = head;
    while (temp != NULL && temp->data != 'P')
        temp = temp->next;

    insertAfter(temp, 'E');
    cout << "After inserting E after P: ";
    traverse(head);

    deleteNode(head, 'C');
    cout << "After deleting C: ";
    traverse(head);

    deleteNode(head, 'P');
    cout << "After deleting P: ";
    traverse(head);

    cout << "Final List: ";
    traverse(head);

    return 0;
}
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

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I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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